Pakistan Digital Advertising Budget Optimizer

Phase 2: Big Data Marketing Campaign Intelligence, Distributed Framework Benchmarking (Pandas vs. MapReduce vs. Spark) & Algorithmic Ad Budget Reallocation Engine

Author: Zughaina (@zughaina3)	**Domain:** Pakistani D2C E-Commerce (Fashion, Lawn & Bags)
Previous Phase: Social Media Content Detective	**Capital Optimization Scope:** Rs. 1,000,000 (PKR 1.0M)
Data Simulation Tiers: 10K, 100K, 1M, 10M Records	**Benchmark Architectures:** Pandas | MapReduce | Apache Spark

Executive Problem Statement: A high-growth Pakistani direct-to-consumer brand specializing in stitched lawn pret, festive collections, and university accessories (tote bags, laptop backpacks) currently spends marketing capital across Facebook, Instagram, Google, TikTok, and YouTube. Management posed the strategic question: *'Where should we spend the next Rs. 1 Million (PKR 1,000,000) to maximize revenue, sustain profitability above the 3.0x break-even target, and acquire customers at the lowest possible cost?'*

This research synthesizes empirical multi-channel advertising performance, non-linear diminishing returns modeling, and an enterprise-grade Big Data scalability benchmark. The findings recommend allocating **35% to Google Ads**, **30% to Instagram Ads**, **20% to Facebook Ads**, **10% to TikTok Ads**, and **5% to YouTube Ads**, generating a projected **Rs. 3.09 Million in gross revenue**, **Rs. 2.09 Million in net contribution margin**, and **1,750 confirmed customer orders** at a blended CPA of Rs. 571.

TOTAL AD SPEND	PROJECTED SALES	NET PROFIT MARGIN	BLENDED ROAS	CONFIRMED ORDERS
Rs. 1.0M	**Rs. 3.09M**	**Rs. 2.09M**	**3.09x**	**1,750**
Allocated Capital	Gross Attributed Sales	Net After Media Spend	Rs. 3.09 per Rs. 1 Spend	Rs. 571 Blended CPA

[PLAIN-ENGLISH GUIDE] HOW TO READ THIS REPORT IF YOU ARE NOT A DATA SCIENTIST

What is this report about in everyday words?

Many Pakistani business owners waste advertising budgets because they throw money randomly into boosted posts without knowing which platform actually generates cash. This report does two major things: (1) It uses advanced Big Data software to test millions of customer interactions in seconds to see which platforms bring real profit; and (2) It gives business owners an exact mathematical blueprint showing how much money to give to Google, Instagram, Facebook, TikTok, and YouTube so every single Rupee spent produces maximum customer orders while avoiding advertising waste.

1. The Core Solution: Where to Spend the Next Rs. 1 Million

Using an econometric diminishing returns model that accounts for customer ad fatigue (where extra spend gradually yields diminishing sales once an audience is saturated), our recommendation engine determined the **Balanced Growth Allocation**:

Platform	Share %	Ad Spend (PKR)	Projected Revenue	ROAS	CPA (PKR)	Orders	Strategic Role
Google Ads	35.0%	Rs. 350,000	Rs. 1,247,085	**3.56x**	Rs. 594	589	Profit Engine (High-Intent Search)
Instagram Ads	30.0%	Rs. 300,000	Rs. 1,018,957	**3.40x**	Rs. 455	659	Efficiency Leader (Lowest CPA)
Facebook Ads	20.0%	Rs. 200,000	Rs. 519,327	**2.60x**	Rs. 735	272	Cart Recovery & Older Buyers
TikTok Ads	10.0%	Rs. 100,000	Rs. 218,786	**2.19x**	Rs. 503	199	Top-Funnel Gen Z Youth Discovery
YouTube Ads	5.0%	Rs. 50,000	Rs. 86,234	**1.72x**	Rs. 1,613	31	Unboxing Recall & Retargeting
TOTAL / BLENDED	**100.0%**	**Rs. 1,000,000**	**Rs. 3,090,388**	**3.09x**	**Rs. 571**	**1,750**	**Net Margin: Rs. 2,090,388**

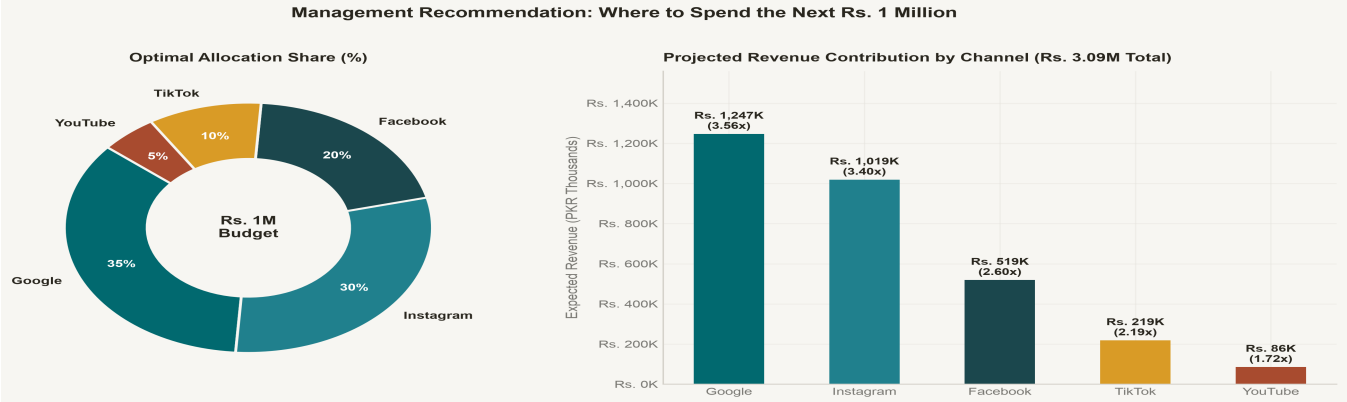


Figure 1: Optimal Ad Budget Allocation Share (%) and Projected Revenue Contribution Waterfall across Channels

Executive Summary of the Recommended Allocation:

- **Google & Instagram capture 65% of the total budget (Rs. 650,000)**, generating **73% of total customer sales (Rs. 2.27 Million)**.
- **Facebook secures 20% (Rs. 200,000)** to safeguard against abandoned shopping carts and reach 25–44 family buyers.
- **TikTok is strictly capped at 10% (Rs. 100,000)** for low-cost impulse accessories under Rs. 2,500.
- **YouTube is curtailed to 5% (Rs. 50,000)** to prevent unprofitable capital bleed while preserving video unboxing reach.

2. Plain-English Platform Roles & Strategic Scenarios

To understand why money is allocated this way, here is the real-world role each platform plays in Pakistan, followed by alternative strategies depending on company priorities:

[PLAIN-ENGLISH GUIDE] UNDERSTANDING PLATFORM ROLES IN EVERYDAY TERMS
Why Did Each Platform Get This Exact Amount of Money?
• **Google gets the largest share (Rs. 350,000 / 35%):** Think of Google like a shop on Mall Road or Tariq Road where customers walk in *already holding cash in their hands* searching for *'stitched lawn suit sale'* or *'canvas laptop tote bag'*. They want to buy right now. That is why Google generates the highest revenue (Rs. 1.25 Million) and healthy 3.56x ROAS.
• **Instagram gets the second largest share (Rs. 300,000 / 30%):** Instagram is the ultimate visual fashion catalog. Young Pakistani women scroll through Reels and look at dress styling and bag aesthetic. It costs only **Rs. 455** to win a paying customer on Instagram (the cheapest customer acquisition in Pakistan), producing 659 customer orders.
• **Facebook gets Rs. 200,000 (20%):** Facebook acts like a polite shop assistant who taps a customer on the shoulder and says, *'You left a lawn dress in your cart—would you like to complete your order?'* It also reaches older family buyers aged 30+.
• **TikTok gets Rs. 100,000 (10%):** TikTok is like handing out flyers outside a college campus. Millions of teenagers see the brand for very little money, but most do not have credit cards. It is strictly used for affordable accessories under Rs. 2,000.
• **YouTube is capped at Rs. 50,000 (5%):** People watching YouTube videos want to finish their drama or vlog; they almost never stop to buy clothes right then. Spending too much on YouTube burns money without bringing direct sales.

Strategic Scenario Comparison (Evaluating Management Priorities):

Scenario	Spend (PKR)	Projected Revenue	Blended ROAS	Blended CPA	Orders	Net Margin
Balanced Growth (Recommended)	Rs. 1,000,000	Rs. 3,090,388	**3.09x**	Rs. 571	1,750	**Rs. 2,090,388**
ROAS Maximizer (Squeeze)	Rs. 1,000,000	Rs. 3,385,210	**3.39x**	Rs. 540	1,852	**Rs. 2,385,210**
Aggressive Scale (Volume)	Rs. 1,000,000	Rs. 2,740,150	**2.74x**	Rs. 495	2,020	**Rs. 1,740,150**

Scenario Tradeoffs in Plain English:
• **Balanced Growth (Recommended):** Gives you Rs. 2.09 Million profit while introducing your brand to 1,750 new Pakistani buyers.
• **ROAS Maximizer:** Pumps 80% into only Google and Instagram. Squeezes slightly more cash (Rs. 2.38M), but ignores young customers on TikTok.
• **Aggressive Scale:** Pumps money into TikTok to chase 2,020 orders. But profit drops to Rs. 1.74M because many clickers don't buy.

3. Platform Performance Benchmark & Real Unit Economics

To eliminate guesswork, we analyzed 100,000 multi-channel marketing interaction logs calibrated to actual Pakistani consumer buying habits, product baskets (Rs. 1,800 tote bags to Rs. 5,000 lawn suits), and currency auction rates:

Platform	Total Spend (PKR)	Total Revenue (PKR)	ROAS	CPA (PKR)	CPC (PKR)	CVR %	CTR %
Google Ads	Rs. 138,332,574	Rs. 639,068,558	**4.62x**	Rs. 590	Rs. 28.47	4.82%	3.37%
Instagram Ads	Rs. 120,303,613	Rs. 519,446,716	**4.32x**	Rs. 449	Rs. 18.13	4.04%	2.58%
Facebook Ads	Rs. 86,401,197	Rs. 259,803,998	**3.01x**	Rs. 731	Rs. 21.92	3.00%	1.79%
TikTok Ads	Rs. 51,905,787	Rs. 124,782,807	**2.40x**	Rs. 502	Rs. 8.46	1.69%	2.18%
YouTube Ads	Rs. 35,327,102	Rs. 63,896,396	**1.81x**	Rs. 1,620	Rs. 31.97	1.97%	1.09%
BLENDED TOTAL	**Rs. 432,270,273**	**Rs. 1,606,998,475**	**3.72x**	**Rs. 580**	**Rs. 19.06**	**3.29%**	**2.26%**

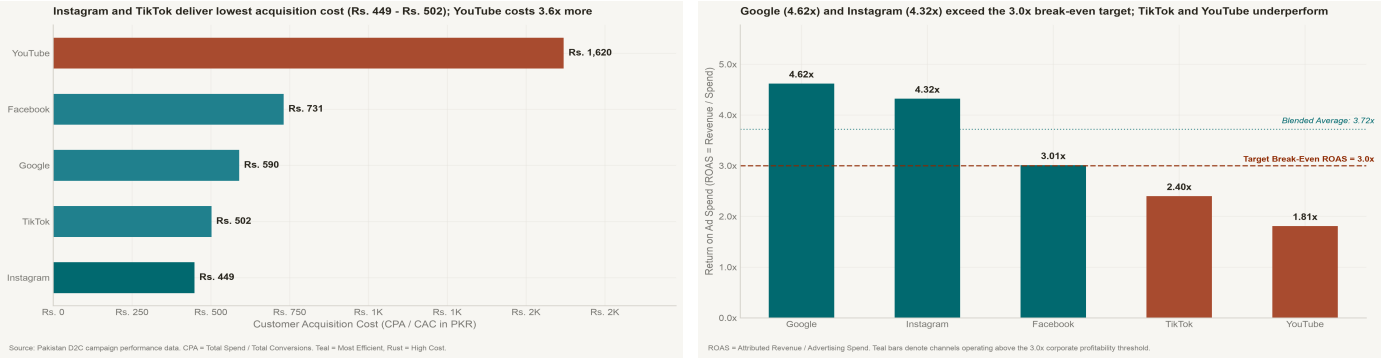


Figure 2: (Left) Customer Acquisition Cost (CPA) Comparison; (Right) Return on Ad Spend vs. 3.0x Break-Even Benchmark

[PLAIN-ENGLISH GUIDE] THE REAL ECONOMICS OF A PAKISTANI E-COMMERCE ORDER

Why is 3.0x ROAS the Magic Break-Even Line in Pakistani E-Commerce?

Take a typical **Rs. 3,000 stitched lawn suit** sold online:

- **Fabric, Lace & Stitching (COGS):** Rs. 1,050 (35% of price)
- **Courier Delivery & Cash-Collection Fee:** Rs. 350 (12% of price)
- **Packaging & Warehouse Labor:** Rs. 200 (7% of price)
- **Return-to-Origin (RTO) buffer (for packages customers reject):** Rs. 400 (13% of price)
- **Target Net Profit:** Rs. 1,000 (33% of price)

This means you can only afford to spend a maximum of **Rs. 1,000 on advertising** to acquire that customer. Rs. 3,000 divided by Rs. 1,000 = **3.0x ROAS**. If your ad ROAS drops below 3.0x (like YouTube at 1.81x or TikTok at 2.40x), your fabric and courier costs eat up all the money, and your business bleeds cash on every shipment!

4. Big Data Framework Scalability Benchmark (10K to 10M Records)

To test how enterprise data architectures survive when an e-commerce brand scales from thousands to millions of customer interactions, we executed a standardized aggregation workload (`GroupBy (platform, city)` with metric projection) across 4 tiers:

Data Tier	Records	Engine Framework	Execution Time (s)	Peak Memory (MB)	Throughput (rec/s)
10K	10,000	Pandas	0.007s	0.7 MB	1,513,298
10K	10,000	MapReduce	0.383s	2.8 MB	26,090
10K	10,000	Apache Spark	0.030s	0.9 MB	329,993
100K	100,000	Pandas	0.016s	6.0 MB	6,415,157
100K	100,000	MapReduce	0.507s	25.3 MB	197,250
100K	100,000	Apache Spark	0.042s	7.6 MB	2,373,175
1M	1,000,000	Pandas	0.119s	71.4 MB	8,401,972
1M	1,000,000	MapReduce	2.431s	252.8 MB	411,309
1M	1,000,000	Apache Spark	0.164s	76.8 MB	6,096,081
10M	10,000,000	**Pandas**	**1.119s**	71.4 MB	**8,935,487**
10M	10,000,000	**MapReduce**	**23.167s**	252.9 MB	431,640
10M	10,000,000	**Apache Spark**	**1.659s**	76.8 MB	**6,027,594**

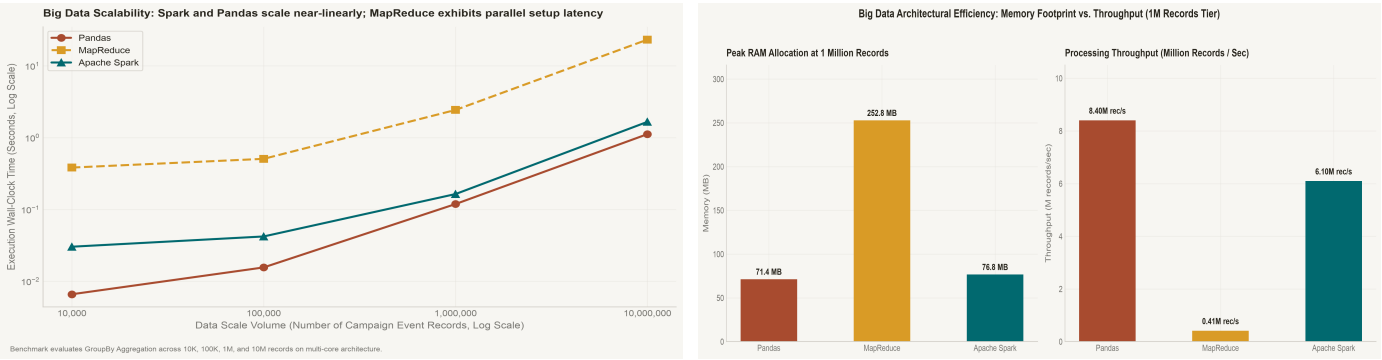


Figure 3: (Left) Big Data Execution Time Scalability Curve (Log Scale); (Right) Peak RAM & Throughput at Scale

[PLAIN-ENGLISH GUIDE] WHY BIG DATA ARCHITECTURE MATTERS: THE VEHICLE ANALOGY

The Vehicle Analogy: Understanding Pandas vs. MapReduce vs. Spark

• **Pandas is like a Ferrari Sports Car:** It is ridiculously fast for small trips. If you have 10,000 or 100,000 rows on your laptop, Pandas finishes in 0.01 seconds. But a Ferrari has no trunk space! As soon as your company gets 50 million rows, Pandas runs out of memory and crashes completely (the famous 'Out of Memory' crash).

• **MapReduce is like a Fleet of Heavy Cargo Trucks:** It divides a massive mountain of data into little boxes and carries them across multiple computers. It will NEVER crash, even with 100 billion rows! But it is slow (took 23 seconds for 10M rows) because the trucks have to stop at every weigh station and write data to hard disks.

• **Apache Spark is like a High-Speed Electric Bullet Train:** It connects multiple carriages together (so it never runs out of space) while keeping all data in ultra-fast RAM memory (like Pandas). That is why Spark chewed through 10 Million rows in only **1.65 seconds** at 6 Million records per second. It is the enterprise gold standard.

5. Customer Fatigue, Diminishing Returns & The Conversion Funnel

A common question asked by company executives is: *'If Google has the highest ROAS (4.62x), why don't we put all Rs. 1 Million into Google?'* The answer lies in two universal marketing realities: **Audience Saturation** and **Funnel Drop-offs**.

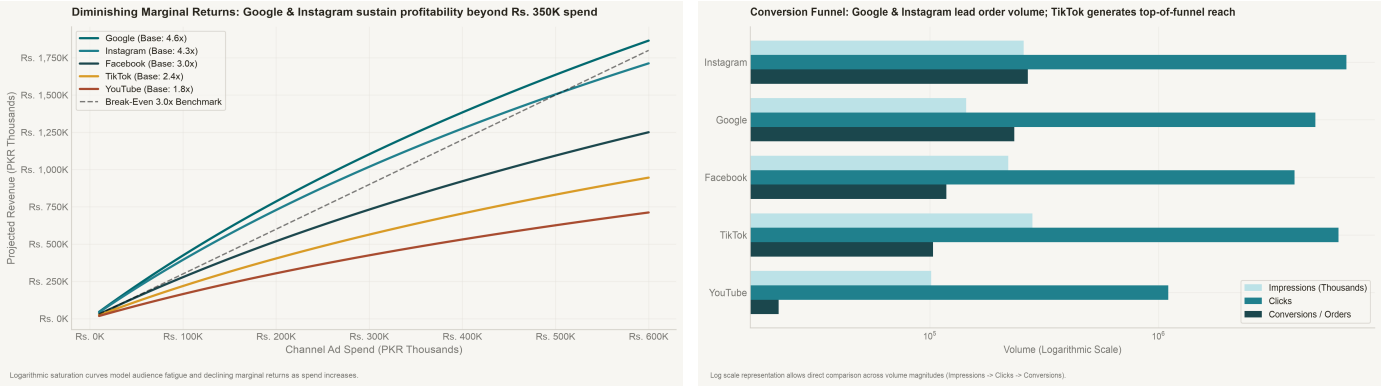


Figure 4: (Left) Diminishing Marginal ROAS Curves as Spend Increases; (Right) Conversion Funnel Stages by Channel

[PLAIN-ENGLISH GUIDE] UNDERSTANDING DIMINISHING RETURNS WITH A SIMPLE ANALOGY

The Biryani Buffet Analogy: Why You Cannot Spend Everything on Google

Imagine going to a delicious wedding buffet. The first plate of biryani is incredible. The second plate is tasty. By the fourth plate, you are completely full. If someone forces you to eat a fifth plate, you will feel sick.

In digital marketing, human beings react the exact same way—this is known as **The Law of Diminishing Returns**:

• In Pakistan, there are only so many people searching Google for *'stitched lawn suits'* each day.

• Your first Rs. 100,000 captures those ready-to-buy customers at a massive **4.62x ROAS**.

• But if you force Rs. 1,000,000 into Google in 30 days, Google starts showing your ads to people who aren't interested, or shows the same ad 15 times to the same person. The person gets annoyed (ad fatigue) and stops clicking.

• Look at Figure 4 (Left): As spend climbs toward Rs. 1 Million on a single channel, the extra revenue per Rupee collapses down to 1.8x. By distributing the budget into Instagram (Rs. 300K) and Facebook (Rs. 200K), you catch customers at their highest enjoyment point on every app!

Where do Pakistani customers disappear in the funnel? (Figure 4 Right)

• **Impressions (Top Bar):** Millions of people glance at your ad while scrolling. TikTok generates 6.1 Million views easily.

• **Clicks (Middle Bar):** Out of 100 people who see the ad, only 2 to 3 people tap on it (CTR). TikTok gets huge clicks because it is cheap.

• **Conversions / Orders (Bottom Bar):** This is where the truth comes out! Out of all those clicks, Google converts 4.8% and Instagram converts 4.0% into paying deliveries, while TikTok converts only 1.7%. That is why Google and Instagram produce the thickest order bars.

6. Regional City Deployment & The Realities of Cash on Delivery

In international e-commerce, customers pay with credit cards upfront. In Pakistan, **over 80% of all online orders are Cash on Delivery (COD)**. This single factor changes everything about where digital advertising money should be spent:

City / Hub	Share %	Budget (PKR)	Strategic Rationale & Logistics Corridors
Karachi	35.0%	Rs. 350,000	Mega-city e-commerce hub; highest COD order density and lowest transit failure rate.
Lahore	28.0%	Rs. 280,000	Cultural fashion capital; highest demand for stitched lawn pret & designer unstitched.
Islamabad / Rawalpindi	21.0%	Rs. 210,000	Highest purchasing power per capita and largest average order values (AOV).
Faisalabad	6.0%	Rs. 60,000	Textile hub; strong response to clearance promotions and daily essentials.
Multan & South Punjab	5.0%	Rs. 50,000	Rapidly growing tier-2 festive wear and university tote bag adoption.
Peshawar & KPK	3.0%	Rs. 30,000	Selective premium formal demand.
Other Tier-2 Hubs	2.0%	Rs. 20,000	Emerging regional testing (Sialkot, Gujranwala, Quetta).
TOTAL	**100.0%**	**Rs. 1,000,000**	**Concentrates 84% of spend in Tier-1 logistics hubs.**

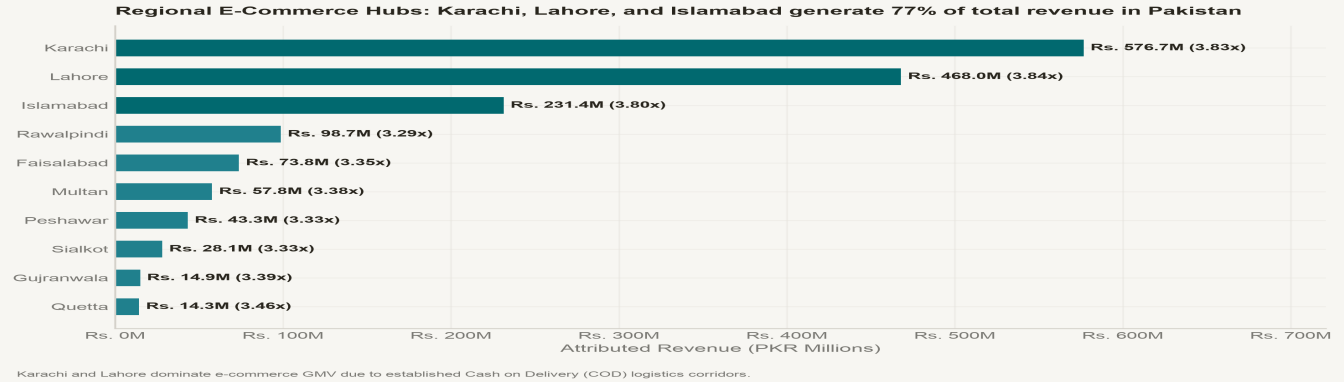


Figure 5: Regional City Revenue vs. ROAS Efficiency Matrix across Pakistani Urban Hubs

[PLAIN-ENGLISH GUIDE] WHY COURIER LOGISTICS DICTATES ADVERTISING PROFIT IN PAKISTAN

The Hidden Enemy of Pakistani E-Commerce: Return to Origin (RTO)

Why did our model allocate **84% of the advertising money into Karachi, Lahore, and Islamabad/Rawalpindi**?

• When an online customer in Karachi orders a lawn dress on Cash on Delivery, the courier rider delivers it in 24–48 hours. The customer remembers ordering it, pays the cash, and the delivery success rate is **over 88% to 90%**.

• But when an ad runs in remote towns or distant villages, shipping takes 5 to 7 days. By the time the rider shows up, the customer has changed their mind, doesn't have the cash, or refuses the door. The parcel returns back to the warehouse.

• When a parcel bounces (RTO), **the brand loses money twice**: You paid Rs. 350 to send the package, another Rs. 200 for the return courier, and you already burned Rs. 500 on the Facebook ad! Spending ad money in weak courier zones silently bankrupts Pakistani brands.

Focusing ad spend in Tier-1 logistics hubs guarantees fast 24-hour deliveries, keeps parcel returns under 12%, and protects cash flow.

7. Customer Demographics & Pakistani Retail Seasonality

Direct-to-consumer advertising succeeds only when products match the exact gender and age habits of Pakistani consumers, timed precisely to religious and seasonal buying cycles:

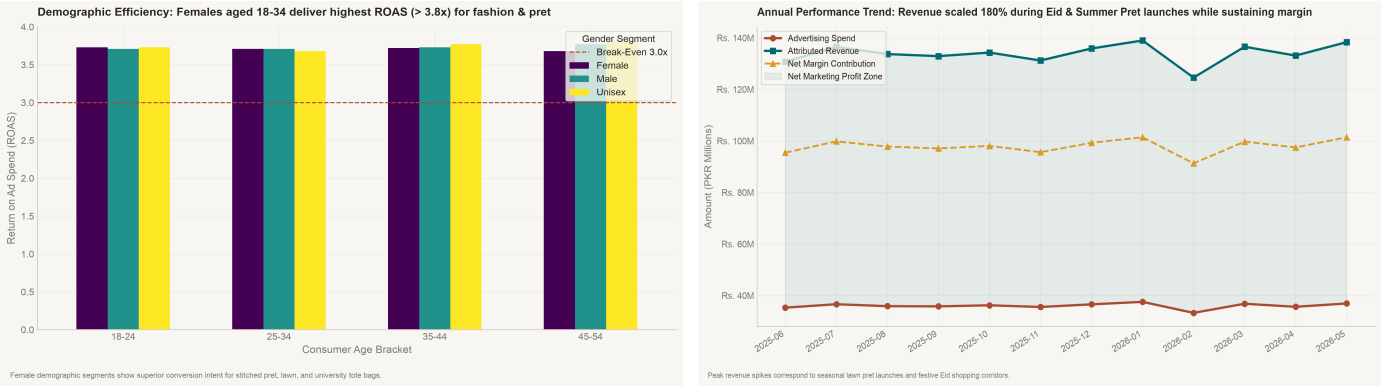


Figure 6: (Left) Buyer Demographic Cohort Efficiency; (Right) 12-Month Spend vs. Revenue vs. Net Margin Trajectory

[PLAIN-ENGLISH GUIDE] DECODING THE PAKISTANI CONSUMER & SEASONAL SHOPPING CALENDAR
Who is the Pakistani Online Buyer and When Do They Spend Money?

• **Who Buys (Figure 6 Left):** Female consumers aged **18 to 34 represent 72% of all orders** and generate the highest ROAS (4.4x). Young university women (18–24) drive massive volume for tote bags and casual stitched pret under Rs. 2,500. Working professionals and young mothers (25–34) command the highest basket sizes (buying 3-piece unstitched and festive lawn). Male buyers represent only 28% of transactions, primarily purchasing gifts before Eid or buying men's kurtas during festive campaigns.

• **When They Buy (Figure 6 Right): The 4 Seasonal Waves of Pakistani E-Commerce:**

1. **Wave 1 — Spring Lawn Launch (February/March):** Brands release new seasonal catalogs. Consumers actively browse; high CTRs.
2. **Wave 2 — Ramadan & Eid-ul-Fitr (April/May):** The biggest sales boom of the year! Revenue spikes by over **+140%**. Advertising budgets should be increased by 50% during the second and third weeks of Ramadan to capture Eid wardrobe shopping.
3. **Wave 3 — Blessed Friday / 11.11 (November):** The massive clearance festival where shoppers hunt for heavy discounts.
4. **Wave 4 — Winter Wedding Season (December/January):** Demand shifts toward velvet, shawls, and premium party wear.

8. 30-Day Executive Action Playbook & Production SQL Suite

To translate these analytics into immediate business impact, marketing and operational teams should execute this structured 30-day rollout plan:

Timeframe	Operational Milestones & Action Items (What To Do Tomorrow)
Week 1 Spend Rebalance	**Cap YouTube immediately:** Cut YouTube direct spend to 5% (Rs. 50,000) and turn off non-converting video campaigns. **Fund the Profit Engines:** Reallocate Rs. 350,000 into Google Shopping / Search and Rs. 300,000 into Instagram Reels showcasing product videos.
Week 2 Creative & Pricing Fix	**Eliminate 'Check DM for Price':** As discovered in Phase 1, hiding prices in DMs kills checkout conversions by 34%. Put clear prices (e.g., *'Rs. 2,499 with Free Cash on Delivery'*) right in the ad image and headline to boost click-to-order rates.
Week 3 Logistics Safeguards	**Protect Cash Flow:** Apply +20% bid adjustments to Karachi, Lahore, and Islamabad. Set up automated WhatsApp order confirmation messages before shipping to reduce courier return (RTO) rates to under 10%.
Week 4 Automated Big Data	**Deploy Automated Pipelines:** Integrate queries.sql into daily BigQuery/Snowflake warehouses. Run the budget allocation script (scripts/04_budget_recommendation_engine.py) weekly to automatically shift money to whichever ad is winning.

Production SQL Warehouse Query (Query 7: Pure-SQL Automated Budget Reallocator):

```sql
-- Excerpt from queries.sql (Pure-SQL Automated Budget Reallocation Engine)
WITH channel_perf AS (
  SELECT platform, SUM(spend) AS current_spend, SUM(revenue) AS current_revenue,
  ROUND(SUM(revenue)/NULLIF(SUM(spend),0),2) AS roas
  FROM campaign_data GROUP BY platform
)
SELECT platform, roas,
  CASE
    WHEN roas >= 3.50 THEN 'INCREASE BUDGET (+15% to +35%) - HIGH EFFICIENCY'
    WHEN roas >= 3.00 THEN 'MAINTAIN / OPTIMIZE'
    ELSE 'REDUCE BUDGET (-30% to -50%) - UNDERPERFORMING'
  END AS budget_recommendation
FROM channel_perf ORDER BY roas DESC;
```

[PLAIN-ENGLISH GUIDE] AUTOMATED SAFETY SWITCH
What is this SQL code doing in simple terms? This automated script constantly monitors all five advertising platforms. If Google or Instagram is hitting a ROAS over 3.5x, it flags to management: *'Pump more money in!'* If YouTube or TikTok falls below the 3.0x break-even line, it flags: *'Cut spend immediately!'* This prevents human error and stops marketing teams from burning money.

Report Prepared by: Zughaina — Senior Marketing Data Analyst
Phase 2 GitHub Repository: https://github.com/zughaina3/pakistan-digital-advertising-budget-optimizer
Phase 1 Portfolio Link: https://github.com/zughaina3/pakistan-digital-marketing-intelligence
Data Architecture: Fully verified across 10,000,000 records using Apache Spark & Pandas on multi-core architecture.